

A Hybridized Approach for Enhanced Fake Review Detection

Shu Xu , Haoqi Cuan , Zhichao Yin , and Chunyong Yin 

Abstract—User reviews on online consumption platforms are crucial for both consumers and merchants, serving as a reference for purchase decisions and product improvement. However, fake reviews can mislead consumers and harm merchant profits and reputation. Developing effective methods for detecting deceptive reviews is crucial to protecting the interests of both parties. In recent years, research on fake review detection has focused on improving machine learning and neural network methods to enhance the accuracy of fake review detection, neglecting the fundamental and necessary work of text feature representation for reviews. High-quality review text feature representation affects or even determines the quality and performance of fake review detection methods. The increasing prevalence of fake reviews results in a more complex distribution within the feature space of review texts, thus necessitating review embedding methods that exhibit comprehensive semantic comprehension and contextual awareness of review texts. To improve the quality of textual feature representation, we propose a review-embedding attention-based long short-term memory (A-LSTM) method that can encode the global semantics of reviews and detect the deception of the review content. A-LSTM uses attention gates to discover the importance of words, and by analyzing the importance of words, it can help distinguish the characteristics of real and fake reviews, and we propose an attention loss function to solve the problem of class imbalance. On the Yelp dataset, the accuracy of deceptive review detection has increased to 90.9%.

Index Terms—Attention, deceptive review detection, feature representation, long short-term memory (LSTM) network.

I. INTRODUCTION

WITH the development and progress of the Internet, various traditional industries have deployed online shopping platforms. User needs drive continuous innovation in consumption patterns and content, and continue to broaden the application scenarios and boundaries of mobile Internet. In 2020, novel coronavirus pneumonia had a huge impact on

the consumer market. Under the influence of the pandemic, consumers' online consumption has significantly increased. The online consumption field has expanded from commodities to medical, education, cultural, and entertainment services and catering. Emerging consumption models such as food delivery and home-to-home services in shopping malls have accelerated their development. The trend of online consumption in the consumer sector shows that people increasingly prefer to use online platforms for shopping, travel, accommodation, and dining. At the same time, people are more inclined to share their consumer experiences on the Internet [1] and consult reviews before making a purchase. According to Truelist's latest online review statistics [2], which were updated on 1 July 2023, approximately 5–10% of the global digital buyer population actively engages in writing reviews, exerting a significant influence on the total revenue generated by e-commerce, contributing to an estimated \$400 billion. Furthermore, just one review on the product page can scale conversion rates by 354%, and globally, 89% of customers read reviews before making a purchase. Taobao, Meituan, Amazon, and Yelp, among other online consumption platforms, host an extensive collection of user reviews related to shopping and consumer experiences. Consumers will refer to other users' reviews before making a purchase decision while merchants can improve products by analyzing user reviews. However, fake review spammers write fraudulent reviews that provide unrealistic advocacy or slander of products or services to influence users' perspectives or consumer behavior [3], greatly harming the interests of both consumers and merchants. On the one hand, deceptive positive reviews can mislead consumers to buy low-quality products, on the other hand, deceptive negative reviews can damage the reputation of merchants and cause damage to their interests. This provides a strong motivation to promote the generation of deceptive reviews [4], [5]. As of 2016, the review site Yelp has more than 108 million reviews [6], and the annual increase in reviews exceeds 18 million [6]. By 2023, Yelp boasts over 244 million reviews [2]. Nevertheless, the prevalence of fake reviews continues to increase. 82% of consumers have reported reading fake reviews in the past year, with 47% believing that companies manipulate their reputation through these fraudulent reviews [2]. Deceptive reviews mislead consumers' shopping decisions, undermine the consumer experience, damage the image of the merchants, undermine the interests of the business, and affect the improvement of the business. Since deceptive reviews have considerable harm, it is important to detect these

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deceptive reviews. The detection of fake reviews, alongside the detection of malicious behavior such as load data manipulation in power systems [7], defense against cross-site scripting (XSS) attacks on smart devices [8], and detection of botnet attacks on Internet of Things (IoT) devices [9], all fall under the category of malicious behavior detection on networks. These efforts are crucial for enhancing the security and stability of cyberspace.

Because each review is a text, detecting deceptive reviews can be viewed as a text classification task [10]. Although natural language processing technology has been widely used in spam detection and spam web detection, it is still a difficult problem to detect deceptive reviews [11]. Current text representation methods have limitations in addressing the complexity of fake review detection. Traditional approaches often fail to effectively capture the semantic information of review texts, resulting in poor performance of detection algorithms in distinguishing between genuine and fake reviews. Generally, deceptive review detection is considered a binary classification problem. Based on the positive and negative samples labeled by people, supervised learning is used to construct a classifier, and then the unlabeled samples are predicted as truthful reviews or deceptive reviews [5], [12]. Review texts are different from news and emails. Most review texts on consumer platforms are short texts and short texts do not always follow the grammatical rules of natural language. They usually contain polysemous words, typos, and even word acronyms, which brings a huge challenge to the correct understanding of text semantics [13]. Compared with spam, its content is obtained by professional counterfeiters imitating real reviews with real shopping experience [14], which is a lie fabricated by the reviewers. The content of deceptive reviews is based on virtual and unreal experiences and opinions. Counterfeiters will imitate reality as much as possible in content, so it is very difficult to distinguish deceptive reviews even for counterfeiters. In the experiment done by Ott et al. [12], the accuracy of manual recognition of three persons was only 57.33%.

Most previous research focuses on improving the accuracy of detection using machine learning and neural network methods. For example, Hajek et al. [15] improve feedforward neural networks to mine sentiment information; Li et al. [16] improve convolutional neural networks to learn document-level representations of reviews; Ren and Ji [17] and Liu et al. [10] improve recurrent neural networks to mine global semantic information of reviews. Shehnepoor et al. [1], [18], Li et al. [19], and Noekhah et al. [20] study graph methods to mine the relationships between reviewers, reviews, and product entities to improve the performance of fake review detection. Aghakhani et al. [21] and Shehnepoor et al. [22] improve generative adversarial networks for review text classification tasks. Ilias et al. [23] evaluate transformer-based fake review detection methods to understand how models make decisions. Budhi and Chiong [24] propose a novel multitype classifier ensemble (MtCE) model with a textual-based featuring method to adequately detect fake reviews. Shunxiang et al. [25] propose a novel fake review detection model based on sentiment intensity and PU learning which structures the review dataset as

streaming data and iteratively improves the model's detection capability. However, this research does not pay attention to the importance of semantic feature mining work for review texts. More data are better than clever algorithms, and good data are better than more data. In real-world scenarios, online reviews exhibit a plethora of lexical items, yet only a select few harbor substantive informational content. The strategic curation of lexemes rich in informational content for text representation has been empirically validated as a means to significantly enhance the efficacy of text classification tasks [26], [27]. Extracting high-quality review text features affects and even determines the quality and performance of fake review detection methods. The representation task of review texts is a basic and necessary work, such as the text similarity of behavioral features is based on the representation of review texts for calculation, graph-based fake review detection methods usually need to represent review texts in feature preprocessing for subsequent knowledge discovery, and in a long period of past research, traditional methods have used review text features to discover the deception of review content. Therefore, a review embedding method that can fully encode review semantic information and improve the performance of fake review detection is very necessary.

Fully mining the semantic information of reviews can effectively help detect fake reviews. The traditional machine learning algorithm adopts the method of feature engineering and artificially designs some criteria to discover, count, and form representative features in the text, including text lexical features and grammatical features. Text lexical features include word frequency, part-of-speech tagging, named entity recognition, stemming, etc. Grammatical features include syntactic analysis, dependencies, location information, etc. Traditional methods have huge human input, ambiguous problems, and difficulties in cross-domain, multitask problems [28]. Feature selection and dimensionality reduction aim to optimize the performance of classification models by reducing the dimensionality of the feature space and selecting the most representative features [26], [27]. However, they may discard important information contained in the original data and even make the model more difficult to interpret. The neural network-based method uses massive text to model text, reflects semantic information, and overcomes tedious feature engineering. When encoding the semantics of reviews, CNN-based methods are similar to N-gram models when used for natural language processing, which makes it difficult to capture features that are larger than N in distance, and important position encoding information is lost during max pooling. Long short-term memory (LSTM)-based methods make it difficult to encode global information based on context due to their linear sequence structure. Bidirectional encoder representations from transformer (BERT)-like methods based on transformer structure mine the semantics of text through pretraining and fine-tuning on specific tasks. However, such methods suffer from many parameters, difficult training, and easy overfitting. Therefore, building an automated and lightweight text feature representation method is a challenge for deceptive review detection.

The increasing number and impact of fake reviews pose substantial losses for consumers and businesses alike. Existing fake review detection technologies are no longer adequate to meet the escalating demand, as the increasing prevalence of fake reviews brings forth a greater diversity of reviews, including those with enhanced concealment techniques and even incomplete reviews. The feature space of review texts exhibits a broader and more intricate distribution, increasing the difficulty of accurately capturing relationships and semantic information between texts. To address this challenge, research in fake review detection requires text representation methods that are more effective and precise. Inspired by prior research, the exploration of more significant terms to capture more informative features effectively enhances detection performance. Therefore, we propose a novel review embedding method. Based on solid experimental datasets and theoretical foundations, we design comprehensive experiments to validate the feasibility of the proposed approach. Our main contributions are as follows.

- 1) We propose an attention-based LSTM (A-LSTM) model to learn the textual representation of reviews, that is, the linguistic features of reviews. By introducing the attention mechanism into LSTM, the model can first learn the more accurate semantic representation of each word in the global semantic environment, and second, the model can also learn the contribution weight of each word to the entire review text. Ultimately, using the review text representation learned by our method to detect deceptive reviews will become more effective. The A-LSTM is a novel LSTM architecture that has not been previously studied, addressing the limitation of sequential execution in LSTM to capture global information, thus achieving stronger representational power.
- 2) Since the deceptive review detection task is a binary classification problem with unbalanced categories, the common loss function makes the model tend to classify all samples into categories with a larger number of samples. We propose an attention loss function to solve this problem. It pays more attention to categories with a small number of samples to balance the bias that occurs during model training, thereby improving detection performance.
- 3) Based on the Yelp dataset collected by Rayana and Akoglu [29], the best detection performance was achieved by the proposed A-LSTM, which improved the accuracy to 90.9%. This confirmed that the review representations embedded by A-LSTM fully encoded the semantic information of reviews, and effectively improved the effect of deceptive review detection. In addition, based on the word contribution weights learned by A-LSTM, the characteristics of deceptive review data and genuine review data were observed, and then the writing techniques of deceptive reviewers were analyzed.

In the remaining parts, Section II presents the related work in deceptive detection. Section III describes the proposed attention-based LSTM. Section IV presents the performance evaluation. Finally, Section V concludes this work.

II. RELATED WORK

In this section, we briefly introduce the related work. They are deceptive review detection, LSTM network, and attention mechanism.

A. Deceptive Review Detection

The research problem of deceptive reviews was first proposed by Jindal and Liu [3], and they divided deceptive reviews into three categories: untruthful opinions, reviews on brands only, and nonreviews. Reviews on brands only and nonreviews are quite different from real reviews, and manual identification is relatively easy. Untruthful opinions are obtained by counterfeiters imitating real reviews, which are confusing and difficult to recognize by humans. Therefore, the research on deceptive review detection focuses on the first type of untruthful opinions. Deceptive reviews are more difficult to identify than spam web pages and spam [30]. For a long time, the difficulty of this research is how to perform more effective feature mining or representation learning on factors such as text, users, and their relationships, so as to improve the accuracy of deceptive review detection.

Initially, researchers extract linguistic features from review content to detect the falsity of the review. Although the lie creator simulates reality as much as possible in content, flaws are still exposed in some details of language and behavior. The bag-of-words feature is a very effective feature for research directions such as opinion mining and sentiment analysis. In the direction of deceptive review detection, its effectiveness is higher than other text features, but there are obvious differences in detection effects in different datasets. For example, Jindal and Liu [3] and Ott et al. [12] use bigram features to detect deceptive reviews. On the dataset constructed by the crowdsourcing platform, the bag-of-words feature bigram can reach an accuracy of 89.6%, but in the real review dataset, only 67.8% is achieved. Feng et al. [31] extract unigram and deep syntactic features from review text. They find that there are certain differences in the syntactic characteristics of deceptive reviews and truthful reviews, such as the number of punctuation marks, the number of function words such as “a,” “the,” and “of,” and the number of first-person pronouns. Li et al. [32] extract part-of-speech features through part-of-speech tagging and frequency statistics to detect deceptive reviews. They find that real reviews and deceptive reviews constructed by crowdsourcing have the following characteristics in terms of part of speech. The former contains more nouns, adjectives, prepositions, qualifiers, and conjunctions, while the latter contains more verbs, adverbs, pronouns, and antecedents.

However, experienced counterfeiters imitate real reviews to write spam reviews, so the accuracy of only using the review text to identify spam reviews is not high. As a result, researchers combine behavioral abnormal information to improve detection accuracy. Mukherjee et al. [33] extract a series of behavior features from the Yelp dataset, which increases the accuracy by about 19%. These behavioral features include content similarity, average length of comments, repeated rating behaviors, extreme rating behaviors, and so on. Wang et al. [34] uses tensor decomposition to represent abnormal behavior features. Tensor

decomposition technology is used to transform each user and product into a corresponding vector representation. The advantage of this method is the vectorized representation of global features, which effectively improves the detection performance. Barbado et al. [35] propose an F3 framework to organize the extraction and characterization of features, which can be used to identify the detection capabilities of each feature. Moreover, the F3 framework introduces a novel definition of social features and makes a detailed classification of related features, thereby improving the efficiency of deceptive review detection.

With the development of the phenomenon of spam reviews and the deepening of research, researchers realize that extracting features cannot encode the semantic information of reviews, so they begin to use neural network methods to learn the document representation of reviews. Semantic representation learning is the abstract representation and generalization of text content and semantics. Jiang et al. [36] consider that users do not have enough time and patience when referring to reviews, so they study review summary generation methods to summarize the semantics of reviews. By identifying important sentences through novel preprocessing strategies and developing solutions for supervised and unsupervised scenarios, respectively, their framework is able to generate comprehensive summaries for reviews. The semantic representation-based model uses semantic representation as a feature to predict the category of reviews by learning semantic representation. Some researchers [10], [16], [17] learn the semantic representation of review texts through neural network models. The robustness of semantic features in domain transfer is better than bag-of-words and part-of-speech features. Ren and Ji [17] experimentally propose to use convolutional neural network (CNN) to extract sentence representations of reviews and use a gated recurrent neural network (GRNN) to learn document-level representations of reviews. GRNN can model discrete information, and complex experiments have confirmed the superiority of neural networks in semantic representation. Li et al. [16] propose the sentence weighted neural network (SWNN) model, which utilizes information gained from part-of-speech and first-person pronoun features to calculate sentence weights for text representation. Liu et al. [10] propose bidirectional long short-term memory by word representation and first-person pronouns embeddings (BiLSTMWF) to represent reviews, which achieves better results than SWNN. Although these methods have achieved good results on crowdsourced datasets, they lack verification on real review datasets. By fine-tuning transformer-based architecture, BERT, on the fake review detection task, Kennedy et al. [37] achieve over 90% accuracy on a widely used deception detection dataset. Moreover, a certain part of the review text is often more relevant to the target task [38]. Ilias et al. [23] evaluate the detection performance of six transformer-based models on deceptive review data and use LIME, a model-agnostic framework for interpretability, to understand the decision-making mechanism of the models.

In addition to mining semantic features from review texts and mining behavioral features from reviewers, some research focuses on improving various networks or algorithms to increase the accuracy of deceptive review detection. Network representation learning is able to learn deep information about

the nodes in the network [39]. The attribute information of the reviewer nodes in the review network graph is the behavioral features of the review. The behavioral features describe the attributes of the review nodes. By combining the attribute information of the nodes and the topology of the network during the learning process, it is able to mine the deep fraudulent behavior information of the reviewers. The following methods employ network representation learning to detect fake reviews. Li et al. [19] propose an end-to-end anti-spam comment (GAS) algorithm based on GCN, which constructs a user-comment-product bipartite graph to mine local semantic features and global context features of spam advertisements comments on Xianyu platform, and uses GCN for end-to-end training and prediction. Shehnepoor et al. [18] consider that previous translational knowledge graph embedding methods have limitations in handling N-1 relations and propose to model review networks as heterogeneous information networks for graph inductive learning, classifying different types of entities separately. The cold start problem and the camouflage problem can be addressed by the proposed Dfraud³. Noekhah et al. [20] consider that previous methods do not fully exploit the existing features and apply feature selection and feature fusion to discover the most effective feature combination. They also propose a heterogeneous graph multiiterative algorithm MGSD, which updates the spamicity of entities based on the spamicity of their neighbors in the iterative process, mining and analyzing the inter- and intra-entity influence relationships, achieving fraudulence prediction of fake reviews. Hajek et al. [15] propose an improved deep feed-forward neural network (DFFNN) model and a novel CNN-based fake detection model and investigate the effects of different word, sentence, and sentiment representations on the performance of fake review detection.

Over the past few years, some researchers have conducted tailored studies addressing the issues in the real-world scenario of fake reviews. Rathore et al. [40] argue that background knowledge in the real world of reviewers and reviews can enhance the effectiveness of detecting fake reviewer groups. They propose a modified PCKMeans semi-supervised clustering algorithm, which incorporates partial background knowledge such as the location of reviewers or the IP addresses from which reviewers post their reviews into the algorithm. Experiments are conducted on a real-world Google Play Store review dataset to validate the effectiveness of the method in improving the clustering performance of fake reviewer groups. Shehnepoor et al. [22] address the problem of lack of high-quality data in the field of fake review detection and propose a generative adversarial network ScoreGAN for fake review detection, which incorporates behavioral features into the generation process to improve the performance of fake review detection. They also use the generated reviews for data augmentation, achieving the same performance on small datasets as on complete datasets. Rout et al. [41] take into consideration that past research often involves extracting varying features from different datasets, making model comparisons challenging. Therefore, their study focuses on evaluating the performance of relevant methods when applying different evaluation criteria on a large-scale complete dataset. They empirically evaluate the performance of graph networks in large-scale dataset deception detection,

TABLE I
SUMMARY OF PREVIOUS STUDIES ON FAKE REVIEW DETECTION

Article	Dataset	Method	Performance	Characteristics
Jindal and Liu [3]	Amazon	Content-based features and LR classifier	AUC = 0.7	The first study of fake review detection, considering duplicate reviews as positive examples. Dataset labels are unreliable.
Ott et al. [12]	Gold-standard opinion spam	Bigram features and SVM classifier	ACC = 0.896	Performs well on crowdsourced datasets, but poorly on real-world datasets.
Feng et al. [31]	Ott et al. [12]	Context free Grammar parse trees	ACC = 0.912	Syntactic differences between fake and real reviews were found.
Mukherjee et al. [33]	Yelp	Behavioral features and SVM	ACC = 0.86	A series of behavioral features are proposed to improve detection performance.
Li et al. [32]	Gold-standard	SAGE	ACC = 0.65	Differences in POS features between the crowdsourced dataset and the real-world dataset were found.
Wang et al. [34]	Yelp	Representation learning and SVM classifier	F1 = 0.892, ACC = 0.899	Learning review representations by tensor decomposition.
Ren and Ji [17]	Gold-standard released by [32]	GRNN	F1 = 0.823, ACC = 0.835	The method performs well on in-domain and cross-domain datasets.
Li et al. [16]	Gold-standard released by [32]	SWNN	F1 = 0.834, ACC = 0.801	Sentence weights were found to better represent the reviews.
Barbado et al. [35]	E-commerce field	Fake feature framework	F1 = 0.82	Be able to find the detection capability of each features.
Li et al. [19]	Xianyu spam opinions	GAS	AUC = 0.9872, F1 = 0.8217	A transductive graph based method. Cannot transfer to unknown data and tasks.
Liu et al. [10]	Gold-standard released by [32]	BiLSTMWF	F1 = 0.876	Embedding POS and first-person pronouns features into word embeddings improves performance.
Kennedy et al. [37]	Yelp	BERT	F1 = 0.9	Despite the good performance, there are too many parameters and truncated long reviews.
Noekhah et al. [20]	Gold-standard opinion spam	MGSD	ACC = 0.93	The method improves detection performance through feature selection and feature fusion, however, it is not experimented on real datasets.
Hajek et al. [15]	Gold-standard released by [32]	DFNN	AUC = 0.951, F1 = 0.896	Use multiple word features to represent reviews in crowdsourced datasets.
Shehnepoor et al. [18]	Yelp	Dfraud ³	F1-Micro = 0.8385, F1-Macro = 0.7087	Be able to deal with cold start problems and camouflage problems.
Rathore et al. [40]	Google Play Store	Modified PCKMeans	ACC = 0.674	Integrate partial background knowledge to detect fake reviewer groups.
Shehnepoor et al. [22]	Yelp and TripAdvisor	ScoreGAN	Yelp: ACC = 0.8476, TripAdvisor: ACC = 0.7726	Better overcome the data scarcity problem, but limited classification performance.
Budhi and Chiong [24]	Hotel, Restaurant, and Doctor	MtCE	Hotel: F1 = 0.8981, Restaurant: F1 = 0.8637, Doctor: F1 = 0.8988	Integrate multiple types of base classifiers based on specific domain knowledge to adapt to different domains.
Shunxiang et al. [25]	Ott et al. [12], YelpZip, YelpChi	SIPUL	Ott [12]: F1 = 0.861, YelpZip: F1 = 0.866, YelpChi: F1 = 0.852	Treat fake review detection as a data stream classification task, iteratively improving model performance.

conducting a practical assessment of both structure-driven and feature-driven review graph models. Budhi and Chiong [24] address the challenge of real-world fake reviews originating from diverse domains by introducing a novel MtCE method, which integrates multiple types of base classifiers based on specific domain knowledge, leveraging the strengths of individual classifiers to compensate for the weaknesses of others, thereby adapting to fake review detection across different domains. Experimental investigations employing different fusion techniques confirm the superiority of MtCEs over single-type classifier ensembles. Shunxiang et al. [25] have discovered that fake review detection exhibits characteristics of data streaming

in real-world scenarios. They innovatively frame fake review detection as a data streaming classification task. A fake review detection model based on sentiment intensity and PU learning (SIPUL) is proposed to better detect fake reviews by dividing the review data into subsets based on sentiment intensity and iteratively training the model to make fake reviews with hidden features lose their hiding in the latest model context during the continuous arrival of data streams.

Table I summarizes the previous fake review detection studies.

The goal of fake review detection is to identify and eliminate the dissemination of false information on the internet, which is

akin to other malicious behavior detection efforts aimed at safeguarding users and systems from potential threats. Deng et al. [7] address the issue of resource constraints in cyber-physical power systems (CPPS) and propose an intelligent detection method for load redistribution attacks. This method not only considers the stealthiness of attack behaviors and resource limitations but also utilizes power load pattern learning techniques to enhance the accuracy and efficiency of attack detection. This is crucial for ensuring the secure operation of networked physical power systems. Chaudhary et al. [8] propose a machine learning attack detection framework for XSS attacks targeting smart devices, providing effective security measures for these devices. By utilizing self-organizing maps (SOMs) to classify XSS attack strings, and generating input vectors based on the changing behavior of attack strings in different input fields of the device's web interface, this framework efficiently detects and defends against XSS attacks. Bojarajulu et al. [9] propose a novel IoT botnet detection method, employing an improved hybrid classifier and a novel hybrid optimization approach, yielding significant results. By combining the information gain model with the hybrid classifier, the accuracy of attack detection is enhanced, and further optimization of the detection system's performance is achieved through the introduction of the SMIE method.

B. LSTM Network

Graves [42] propose an LSTM network, which controls the change of cell information through three gating units. LSTM solves the problems of gradient disappearance and gradient explosion during ordinary RNN training [43], so that it can deal with the relationship dependence problem of long sequence data. It has been widely used in text generation, machine translation, and speech recognition. Subsequently, many LSTM variants appear. Gers et al. [44] add "peephole connections" to the structure of LSTM, which allows each gate structure to see cell information and can enhance the network's ability to distinguish subtle features between input sequences. The LSTMs studied above only predict the output at the next time based on the timing information of the previous time. However, in some problems, the output at the current time is not only related to the previous state but may also be related to the future state. Graves et al. [45] propose a bidirectional LSTM, which is composed of two LSTMs in opposite directions superimposed on top of each other. The output at each moment is determined by the information in both directions, which can better encode the context of the text. Cho et al. [46] simplify the LSTM architecture, called the gate recurrent unit (GRU), the basic structure consists of a reset gate and an update gate. Compared with the LSTM architecture, GRU has fewer parameters, and its training speed is faster when the amount of data is large. Xu et al. [47] introduce an attention mechanism to the problem of image caption, which will select useful parts of the image to get useful input, so that RNN can get more effective output. Merity et al. [48] use DropConnect and average stochastic gradient descent optimization strategy to address the overfitting issue during LSTM training, thereby enhancing the performance of the task.

The ON-LSTM proposed by Shen et al. [49] utilizes neuron ordering to represent the hierarchy of information, thereby embedding hierarchical structure to learn the syntactic structure of sentences. Melis et al. [50] propose the mogrifier LSTM, which introduces additional gating operations to enable multiple rounds of interaction computation between the input x and the hidden state h before inputting them into the LSTM. This ultimately enriches the interaction representation between the input and the context. However, these studies consistently overlook the limitation of sequential execution in LSTM, which fails to capture global information during the computation process.

C. Attention Mechanism

The attention mechanism is a special structure that people embedded in the machine learning model to automatically learn and calculate the contribution of input data to output data. Mnih et al. [51] use the attention mechanism on the RNN model to classify images and learn the part of an image to be processed through attention. Subsequently, in the field of natural language processing, Bahdanau et al. [52] apply the attention mechanism to neural network machine translation. Luong et al. [53] also propose a global attention mechanism and a local attention mechanism for machine translation problems. Vaswani et al. [54] propose a transformer network that completely abandons the traditional convolutional neural network and recurrent neural network, and the entire network structure is completely composed of the attention mechanism. But it also has its own limitations. The transformer can only handle fixed-length text strings. Before entering the system, the text must be divided into a certain number of paragraphs or blocks, this kind of text block will lead to context fragmentation [55]. BERT [56] use a multilayer bidirectional transformer encoder and successfully achieve state-of-the-art results in 11 NLP tasks at the time of proposal. ELECTRA [57] proposes a new generative-discriminative pretraining task called replaced token detection, which replaces some tokens in the original text with generator outputs, and predicts whether the tokens have been replaced by the discriminator. Compared with BERT, ELECTRA achieves higher computational efficiency and better performance. Decoding-enhanced BERT (DeBERTa) [58] introduces a disentangled attention mechanism that separately encodes content and position information and an enhanced decoding mechanism that improves the quality of the generated text by incorporating the disentangled attention mechanism. The efficiency of model pretraining is enhanced by improving BERT.

III. THE PROPOSED MODEL

In this section, we will introduce our proposed model for detecting deceptive reviews. In order to detect deceptive reviews, we consider learning the document-level representation of the review text to obtain the linguistic features of the review. The model prioritizes mining the importance of words to capture crucial linguistic features for identifying deceptive reviews, aligning with the core concept of feature selection

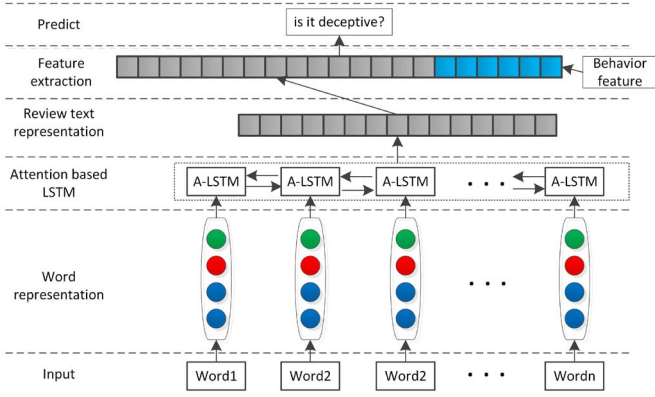


Fig. 1. Architecture of deceptive review detection model.

and dimensionality reduction, aiming to enhance the classification model's performance by selecting the most informative features. The importance of a word directly influences its prominence in the review's representation. Hence, the document-level representation should emphasize the semantics of these crucial words and sentences, given that deceptive reviews often contain exaggerated language or extreme sentiments confined to specific phrases. To address this, we propose the A-LSTM for learning the document-level representation of review text. By incorporating the attention mechanism, our model dynamically assigns significance to different segments of the review, allowing for a nuanced understanding of each word's contribution. By assigning greater importance to deceptive words, document-level representations can effectively encapsulate the semantic nuances associated with deceptive language. Furthermore, we integrate behavioral features of reviewers with the document-level representations to extract comprehensive features for detecting deceptive reviews. The overall structure of our proposed deceptive review detection model is depicted in Fig. 1.

As shown in Fig. 1, the model first segment the review text into words, and then extract the word representation of each word as the input of the A-LSTM. The A-LSTM takes these word representations as input, capturing the semantic information of the current word in the global context during sequential computation. It mines the importance of the current word and further refines and denoises the semantics of words through an attention layer. Based on the importance of words, the word representations are aggregated to obtain the semantic representation of the review, i.e., the linguistic feature of the review. In addition to linguistic features, the behavioral features of reviewers are also extracted to assist in detecting deceptive reviews. Then, the obtained review text representation vector and the behavior feature vector are concatenated and used as the input of the deceptive review detection classifier to predict whether the review is a deceptive review.

A. A-LSTM

In this section, we introduce our text representation learning model A-LSTM in detail. The input of A-LSTM is the word

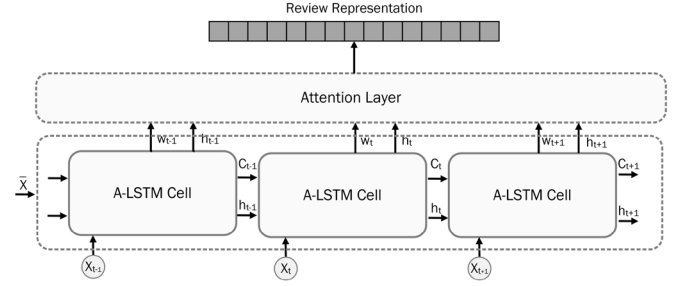


Fig. 2. Structure of attention-based LSTM.

representation of each word, and the output is the document representation of the review. The overall structure is shown in Fig. 2.

As shown in Fig. 2, A-LSTM consists of two layers in total. The first layer takes word representations as input, capturing the semantic importance of words and the global semantic features of the words. The second layer further refines the representations, learning long-distance dependencies, and aggregates word representations based on their importance, generating the final representation of the review. The detailed explanation of these two layers is as follows.

The first layer is similar to the classic LSTM. The LSTM neuron transmits two pieces of information backward in the time dimension, namely the cell state and the hidden state. The hidden state is obtained after the cell state passes through a neuron and an output gate, so the memory contained in the hidden state is actually the content after the cell state decays. In addition, the cell state is transmitted along the time axis in a channel with less attenuation, and the ability to retain information with a larger time span is much stronger than the hidden state. Therefore, what is stored in the hidden state is mainly recent memory, and What is stored in the cell state is mainly long-term memory. The detailed calculation process of the A-LSTM cell can be seen from (1)–(8). The A-LSTM cell has three gating units, namely, forget gate, input gate, and output gate, which are consistent with classic LSTM. The forget gate is responsible for discarding past cell state information, that is, long-term memory. The input gate is to determine what new information is stored. The output gate determines how much recent memory is output. They are calculated as shown in the following equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o). \quad (3)$$

LSTM's sequential execution limits its ability to incorporate global information and encode input comprehensively. To address this limitation and enrich long-term memory with global insights, we introduced an attention mechanism in A-LSTM. This mechanism facilitates the incorporation of new information into the cell from a global perspective. The calculation of attention state information, as depicted in the following

equation, further enhances the model's capability to capture the broader context and refine its representations accordingly:

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (4)$$

$$a_t = \tanh(W_a \cdot [\tilde{c}_t, \bar{x}] + b_a). \quad (5)$$

$\tilde{c}_t$ serves as the candidate cell state, while $\bar{x}$ represents the rough global semantic information. By combining the current candidate cell state with the rough global semantic information, we derive the attention state information. This information encapsulates both the new insights from the current time step, represented by the candidate cell state, and the broader semantic context of the entire text environment. Consequently, the attention state information effectively captures the current cell state within the global context. Subsequently, the value of the input gate, as indicated in the following equation, determines the degree to which attention state information is integrated into the long-term memory. Given that the attention state information embodies global insights, the long-term memory c_t is enriched with comprehensive semantic information

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t + (1 - i_t) * a_t. \quad (6)$$

$\tilde{c}_t$ represents the candidate cell state, while $\bar{x}$ signifies the rough global semantic information. By combining the current candidate cell state and the rough global semantic information, we obtain the attention state information. Since the candidate cell state encapsulates new information at the current time step and the global semantic information encompasses the semantics of the entire text environment, the attention state information effectively captures the current cell state within the broader global context. Subsequently, leveraging the value of the input gate, as indicated in (6), determines the extent to which attention state information is integrated into the long-term memory. Consequently, since the attention state information contains global insights, it enriches the long-term memory c_t with global semantic information

$$w_t = x_t \cdot W_{\text{attention}} \cdot c_t^T \quad (7)$$

$$h_t = o_t * \tanh(c_t) \quad (8)$$

$$w_t = x_t U \Sigma(c_t V)^T. \quad (9)$$

The row vector of U represents the attention information corresponding to each dimension of the input. The row vector of V represents the attention information corresponding to each dimension of long-term memory information. Then the $x_t U$ calculation can get the input attention information and $c_t V$ can get attention information of long-term memory. Σ can be understood as the degree of correlation between the input word vector attention information and the corresponding dimensions of the attention information of long-term memory. The larger the value, the more important. In this way, the learned $W_{\text{attention}}$ can represent the correlation between the input word vector and long-term memory. The larger the w_t , the stronger the semantic relevance of the input word vector at this time step to the sentence in the past time step, and the greater the importance of this word.

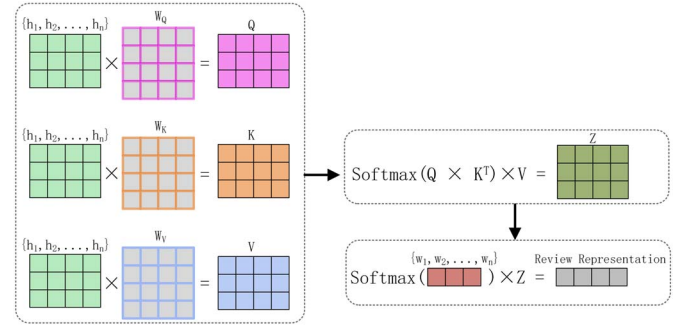


Fig. 3. Calculation process of the attention layer.

As shown in (8), the hidden state h_t is calculated through the output gate and the cell state c_t . As mentioned above, we added global information to c_t , so the output value h_t also adds global information. We regard h_t as a more accurate semantic representation of words in the global semantic environment. However, the first layer of A-LSTM is executed sequentially. Although h_t adds global semantics, its expressive power is still limited and lacks the remote dependence of learning individual words. So we will further enrich the semantic representation of each word in the second attention layer.

In summary, at time step t , the input of A-LSTM cell is $x_t, \bar{x}, c_{t-1}, h_{t-1}$. $\bar{x}$ is a rough global semantic representation. c_{t-1} is the cell state information of the previous time step. h_{t-1} is the output information of the previous time step. The output of the A-LSTM cell at this time step is c_t, w_t, h_t . From (6)–(8), it can be seen the detailed calculation process. They are respectively the new cell state at this time step, the importance of the input word to the global semantics, and the output information of the time step. Since the long-term memory of A-LSTM contains global information, the output h_t of each step can be regarded as the semantic representation of the word at time t in the global semantic environment. On the whole, assuming that there are n words in the review text after word segmentation, the input of the first layer is $x = x_1, x_2, \dots, x_n$, and $\bar{x}$ is a rough word representation. The output is $w = w_1, w_2, \dots, w_n$ and $h = h_1, h_2, \dots, h_n$. w is the importance of each word, and h is the semantic representation of each word in the global semantic environment. After the first layer of A-LSTM, we get a more accurate semantic representation of each word. Next, w and h will be used as the input of the attention layer to calculate the text representation. The attention layer will further optimize the semantic representation of each word in the global environment.

The second layer is the attention layer, where the input is $w = w_1, w_2, \dots, w_n$ and $h = h_1, h_2, \dots, h_n$, and the output is the review representation rr . The main purpose of this layer is to learn the long-term dependency relationships between words and aggregate word representations into a comment representation based on attention weights. The calculation process of the attention layer is shown in Fig. 3.

As shown in Fig. 3, h is the more accurate semantic representation of each word obtained after the review text passes through the first layer of A-LSTM. However, it lacks the learning of distance dependence between words. At this level, we learn

the long-distance dependence between each word and obtain a richer semantic representation. The attention layer can be described as mapping a query and a set of key-value pairs to an output. The output is calculated as a weighted sum of values, where the weight assigned to each value is calculated by the compatibility function of the query and the corresponding key. The query, key, and value are respectively calculated by h , and we pack them into matrices Q , K , and V . Then Q , K , and V are calculated by the following equation to get the output matrix Z . Formula (10) means that we match the key according to the query, and then get the matched content according to the similarity. Because Z is good modeling the dependencies between each word, so it can be a good representation of the semantics of each word in the global context environment

$$Z = \text{Softmax}(Q \times K^T) \times V. \quad (10)$$

Since the weight value w calculated in the first layer of weight indicates the importance of each word to the semantic representation of the text, and Z indicates the semantic representation of each word, we weight the semantic representation of the word according to the importance of the word to obtain the final review text representation. The greater the weight, the more important the word, and the greater its contribution to the semantic representation of the text. As shown in the following equation, $w = w_1, w_2, \dots, w_n$ after softmax function does the matrix multiplication calculation with Z to get the review representation rr :

$$rr = \text{Softmax}(w) \times Z, w = \{w_1, w_2, \dots, w_n\}. \quad (11)$$

Although the one-way A-LSTM can already learn the global semantic context information of each word. However, there is a problem in the calculation of the importance of each word, that is, the weight value w . The calculation of the weight value w is only related to the current input and the past cell state information, that is long-term memory. The long-term memory becomes more and more abundant with the continuous calculation of A-LSTM, which results in a difference between the weight value of the previous time step and the weight value of the later time step. Therefore, we use bi-directional A-LSTM to represent text. The forward A-LSTM has a stronger ability to learn weights in the lower part of the review, and the backward A-LSTM has a stronger ability to learn weights in the upper half of the review. We add the text representations in the two directions as the final text representation. This is to balance the difference in the numerical calculation of weights and obtain text features with stronger semantic representation capabilities. Furthermore, we have stacked LSTM layers. Stacking LSTM layers can enhance the model's representational power and learning ability. By increasing the model's depth, it improves its ability to abstract data, adapts better to complex data distributions by increasing the number of parameters, and enables each LSTM layer to learn different time-scale features, thereby enhancing semantic modeling capabilities.

In summary, the A-LSTM model proposed in this article generates a review representation for the review text, the input is the word representation of the word tokenized review text, and the output is the review text representation. The introduction of

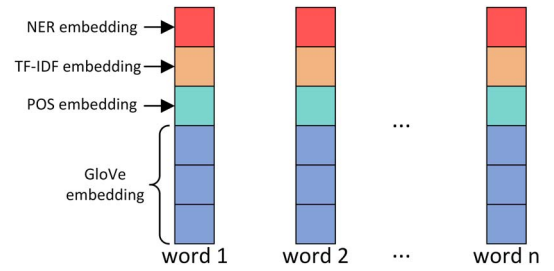


Fig. 4. Composition of word representation.

the attention state in the A-LSTM cell, the weight value w in the A-LSTM cell, and the introduction of the attention layer enable the model to learn the attention degree to each word of review representation. Therefore, compared with manually extracting linguistic features, the semantic expression of the review representation is more accurate in the deceptive review detection task, which improves the detection efficiency.

B. Feature Extraction

In this section, we introduce our word representation feature and reviewer behavior of review feature extraction for review feature extraction.

As shown in Fig. 4, word representation in our proposed model consists of four parts, namely GloVe embedding, part of speech (POS) embedding, term frequency–inverse document frequency (TF-IDF) embedding, and named entity recognition (NER) embedding. These four parts are concatenated to represent words so that A-LSTM can get more semantically rich input. GloVe embedding is a word embedding method pretrained by the global vectors for word representation (GloVe) algorithm [59]. It contains 50-D, 100-D, 200-D, and 300-D versions, through which we can easily obtain the semantic representation of words. However, the corpus used by the pretrained GloVe embedding comes from Wikipedia or Twitter, which is very different from the review text data and cannot represent the meaning of words in the deceptive review detection task. So we use other features to supplement the semantics and grammar of expressing words. POS embedding is the part of the speech feature we extract for each word. Li et al. [32] found that the POS features of fake reviews differed significantly from those of credible reviews, with the former containing more nouns, adjectives, prepositions, and conjunctions, so we extracted the POS features of each word to help the model encode semantics. The existence of POS helps semantic analysis. Mukherjee et al. [60] found that fake reviewers formed fake review groups and fabricated fake reviews. Fake reviewers create fake reviews with high similarity in content because they lack real shopping experience and imitate each other's writing style. These highly similar sections contain words that are not commonly found in credible reviews, so some words in fake reviews are more frequent than in credible reviews. Counting the frequency of all words in the review set for each review can help identify fake reviews. TF-IDF is a classic one-hot encoding method that can evaluate the importance of a word in a document set. In our

model, the TF-IDF value of each word in each review is considered as the TF-IDF embedding of the word. For the fake review detection task, the semantics of some entity words such as place names, product categories, and person names are irrelevant to the semantic representation of the final review. These words are unimportant and we need to flag these words and let the model give less weight to these words when encoding the semantics. NER embedding is the encoding of the entity category of each word, which helps the A-LSTM model encode the semantics of reviews. From this, we get rough word representation features.

A-LSTM needs to capture the information in the global semantic environment of each word, so A-LSTM introduces an attention state. As is shown in (5), the calculation of the attention state requires a global semantic information representation as input. However, the global semantic representation is exactly the text representation we want, and this constitutes an endless loop. Our purpose is to learn the textual representation of reviews, but the model needs the semantic representation of reviews as input to learn the representation of reviews. Therefore, we use rough global semantic representation as input to learn the semantic representation of reviews with stronger semantic expression ability just as we use rough word representations in the first layer of A-LSTM to learn word representations in the global environment. Since the abovementioned rough word representation features have good expressive ability, the article uses the mean value of these word representation features to represent a rough global semantic representation. The calculation formula is as follows, where N is the length of the text, x_n is the above-mentioned word representation feature and $\bar{x}$ is the rough global semantic representation of the text.

$$\bar{x} = \frac{1}{N} \sum_n x_n. \quad (12)$$

Behavioral features are some abnormal behavioral information of review publishers, which can assist the review representation to detect deceptive reviews. In our model, the following behavioral features are mainly extracted. The average content similarity, is the average of the pairwise similarity of all reviews posted by the reviewer. For deceptive review publishers, the time cost of fabricating a new review is high each time, and the time cost of copying the original review of similar products is low. The mutual copying of deceptive reviews is a common method for deceptive review publishers. The content similarity calculation for the reviews posted by the same reviewer can reflect the review behavior characteristics of the reviewer. In our experiment, the average content similarity feature calculation is shown in the following equation:

$$f_{\text{average_similarity}} = \frac{\sum_{i,j} \cos(r_i, r_j)}{m(m-1)/2}. \quad (13)$$

m is the total number of reviews posted by the reviewer. r_i and r_j are two different reviews posted by the reviewer. Average review length, which is the average length of all reviews posted by the reviewer. From the perspective of psychology, people are unwilling to spend too much time on nonspontaneous behavior. Therefore, the length of deceptive reviews is generally shorter than that of truthful reviews. Extreme rating

behavior means that the reviewer has scored the highest or lowest scores in the rating system, and there is the possibility that the reviewer deliberately praised or discredited the product. The positive rating ratio is the proportion of positive reviews in all reviews of the reviewer. Research [33] shows that most of the reviews posted by deceptive review publishers are positive sentiment reviews, while the distribution of sentiment polarity in the reviews of normal reviewers is more balanced. Rating deviation is the difference between the rating of a review and the average score of the product being reviewed. The larger the rating deviation, the more deviation the rating of the reviewer deviates from the public's rating of this product. The average rating is the average rating of all reviews posted by the reviewer. The average rating of normal reviewers is not much different, while the average rating of deceptive review spammer is more extreme. The standard deviation of the rating is the standard deviation of the ratings of all reviews posted by the reviewer. Since deceptive reviews are posted to praise or discredit the product, the ratings of deceptive reviewers are more extreme, so compared with normal reviewers, the standard deviation of the rating of deceptive review publishers is larger. The maximum number of reviews posted on a day, is the maximum number of reviews posted by a deceptive review spammer on a day. Deceptive review spammer will submit a large number of reviews within a day out of interest, which is an abnormal behavior.

C. Attention Loss Function

Deceptive review detection is a binary classification task where truthful reviews significantly outnumber deceptive ones in real-world datasets such as Yelp, resulting in imbalanced sample categories. This imbalance poses challenges for classification models trained with cross-entropy loss, as they tend to favor majority classes, often leading to all samples being classified as truthful reviews.

To address this issue, we propose attention loss. Initially, we quantify the classification difficulty of each sample using (14). Here, $\hat{y}$ represents the model's predicted probability, y denotes the true label, and g is proportionate to the detection difficulty. A higher g indicates greater difficulty in classification. Given the model's inclination to classify data into categories with more samples, truthful reviews (which are abundant) tend to have smaller g values and are easier to classify. Conversely, deceptive reviews, being less frequent, often have larger g values, indicating that they are harder to classify.

Moreover, both deceptive and truthful reviews may contain outliers, further complicating classification. Additionally, irrelevant reviews can negatively impact model training. To mitigate biases toward easy-to-classify samples during model training, we prioritize nonoutlier, difficult-to-classify samples by assigning them higher weights while reducing the weights of easy-to-classify samples. Furthermore, we assign smaller weights to particularly challenging samples, such as outliers, to minimize their influence on model learning

$$g = \begin{cases} 1 - \hat{y}, & \text{if } y = 1 \\ \hat{y}, & \text{if } y = 0. \end{cases} \quad (14)$$

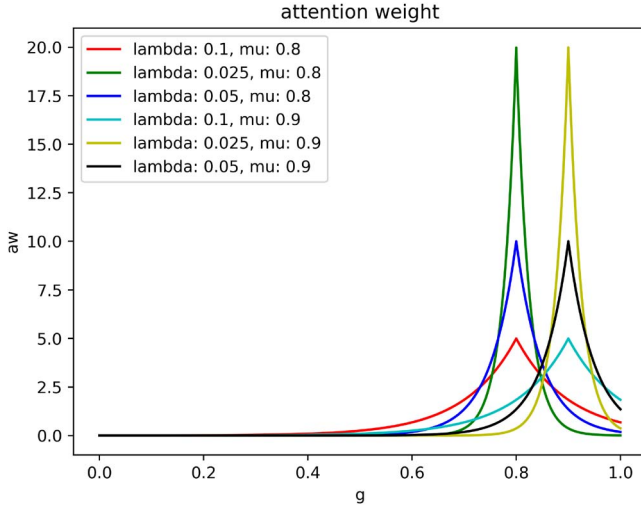


Fig. 5. Weights of the attention loss function for different λ and μ .

Attention loss adopts the strategy of weighting the cross entropy loss function to balance the bias of the model. Specifically, the calculation of the attention weight value is shown in (15), g is the evaluation of the degree of difficulty of the sample in the above part, and λ and μ are hyperparameters. On the one hand, it can be seen from Fig. 5 that when g is equal to μ , the attention weight is the highest. When g is less than μ , the attention weight value keeps increasing with the increase of classification difficulty g . When g is greater than μ , the attention weight decreases continuously. The greater the attention weight, the greater the degree of attention to the sample. μ can be regarded as a threshold to distinguish the outliers. When the degree of difficulty of classification g is greater than the threshold μ , it is considered an outlier, and its weight should be reduced to reduce the degree of attention of the model. And when the degree of greater than the threshold is greater, the degree of outlier is considered to be greater, and a smaller weight is given to it. That is, for outliers, the greater the degree of abnormality, the less attention to them. When the degree of difficulty of classification g is less than μ , we consider it to be a normal sample point, and the attention weight becomes larger as the degree of difficulty of classification g becomes larger. That is, for normal points, the more difficult it is to classify, the more attention should be paid. On the other hand, it can be seen from Fig. 5 that the smaller the λ , the thinner the attention weight function. This means that the ratio of the attention intensity of the nonoutlier difficult-to-classify samples to the attention intensity of other samples is larger. When the λ is larger, the attention weight function is rounder, which means that the ratio of the attention intensity of the nonoutlier difficult-to-classify samples to the attention intensity of other samples is smaller. In summary, the setting of μ is used as a threshold to distinguish outliers, and the setting of λ is used to adjust the intensity of weighting the sample.

$$aw(g) = \frac{1}{2\lambda} e^{-\frac{|g-\mu|}{\lambda}}. \quad (15)$$

TABLE II
EXAMPLE OF THE YELP DATASET

Field	Example
User_id	wKW9oqOU259KkyJVxdsBqQ
Prod_id	Toast
Rating	1.0
Label	1
Date	16 November 2014
Review_text	Drinks were bad, the hot chocolate...

By adding attention weight to the cross-entropy loss, the model can pay more attention to the samples that are difficult to classify, thereby reducing the bias caused by the model training due to the imbalance of the training sample categories. The final attention loss is shown in following equation:

$$L_{al} = \sum aw \cdot (-y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})). \quad (16)$$

IV. PERFORMANCE EVALUATION

A. Experimental Data Set

The Yelp dataset used in our experiment comes from the integration of YelpChi, YelpNYC, and YelpZip collected by Rayana [29]. We removed duplicate reviews from YelpChi, YelpNYC, and YelpZip, and finally got 675 853 reviews. Among them, there are 586 473 truthful reviews and 89 380 deceptive reviews. This dataset contains reviews from 5044 restaurants by 260 277 reviewers in New York and reviews from 201 hotels and restaurants by 38 063 reviewers in Chicago. Each sample of dataset contains the following fields, user_id, prod_id, rating, label, date, and review_text. They are the user's ID, the product's ID, the rating of the product being reviewed, the label of the review, the date the review was published, and the text of the review. A label of 0 represents a truthful review and a label of 1 represents a deceptive review. Table II shows an example of the dataset. We shuffle the order of the samples in the dataset, 90% as the training set and 10% as the test set.

Both the gold standard dataset and the Amazon review dataset are datasets that are often used for fake review studies. The gold standard dataset is a crowdsourced dataset in which fake reviews are obtained from writers imitating real reviews. It has a large gap with real-world reviews and cannot extract behavioral features. Moreover, it has a small amount of data that does not allow the relevant methods to be adequately trained. The Amazon reviews dataset is a nonstandard fake reviews dataset, in which the reviews are not clearly labeled as fake or trustworthy. Moreover, the dataset lacks the reviewer field and the review date field, resulting in some behavioral features such as the positive rating ratio, rating deviation, and the maximum number of reviews posted on a day cannot be computed. Considering the above issues, we do not use the gold standard dataset and the Amazon review dataset in the experimental part.

B. Evaluation Standard

For data classification with unbalanced categories, we need multiple evaluation criteria. In our experiment, we used the

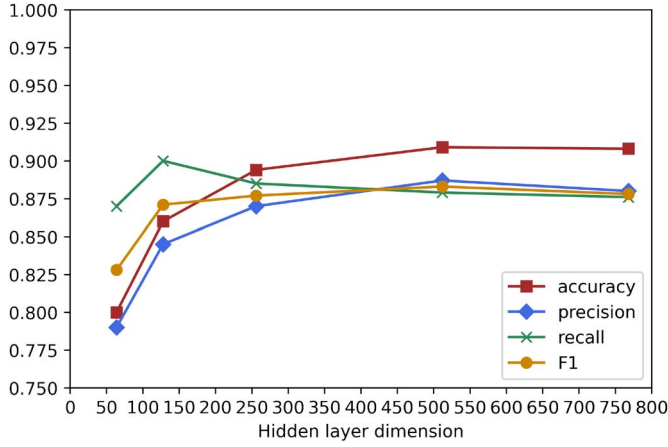


Fig. 6. A-LSTM performance in different hidden layer dimension.

accuracy (ACC), precision (P), recall (R), and $F1$ value. The accuracy is the proportion of correct predictions in the predicted sample. The precision represents the percentage of correct predictions in the sample whose predictions are positive. The recall represents the proportion of all positive samples that are correctly predicted. $F1$ is used to weigh precision and recall, so it is a comprehensive indicator. Their calculation formula is as follows:

$$ACC = \frac{TP + TN}{TP + FP + FN + TN} \quad (17)$$

$$P = \frac{TP}{TP + FP} \quad (18)$$

$$R = \frac{TP}{TP + FN} \quad (19)$$

$$F1 = \frac{2PR}{P + R} \quad (20)$$

TP indicates the number of deceptive reviews correctly classified and FP indicates the number of true reviews that are incorrectly identified while the number of true reviews correctly classified as true reviews is represented by TN and FN represents the number of deceptive reviews that are misidentified as true reviews.

C. Parameter Sensitivity

In the experiment, we tried to set up different hidden layer dimensions, GloVe embedding dimensions and review representation dimensions to study the influence of these parameters on A-LSTM in the detection of fake reviews. Figs. 6–8, respectively show the influence of hidden layer dimension, GloVe embedding dimension, and review representation dimension on A-LSTM when other parameters are fixed. Our experiment tested the performance of A-LSTM when the hidden layer dimensions were set to 64, 128, 256, 512, and 768. It can be seen from Fig. 6 that the overall performance of A-LSTM gradually increases with the increase of the hidden layer dimension, and gradually stabilizes after the dimension is 256. It is worth noting that the recall increases at the beginning and then decreases,

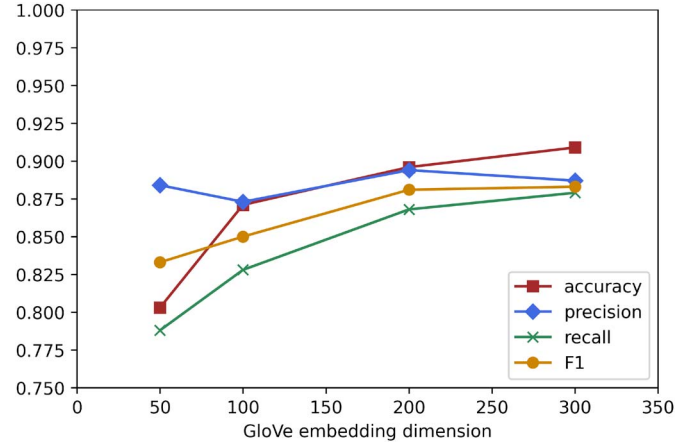


Fig. 7. A-LSTM performance in different GloVe embedding dimension.

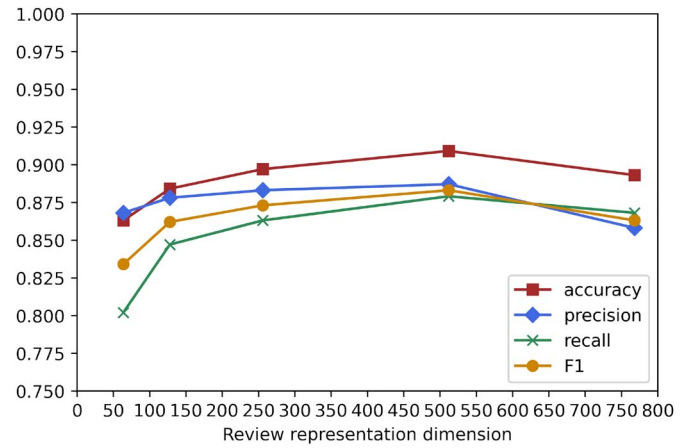


Fig. 8. A-LSTM performance in different review representation dimension.

which indicates that too rich expression ability of the semantic representation obtained by the first layer of A-LSTM will reduce the recall of the model. When the dimension reaches 512, the overall performance hardly changes. Fig. 7 shows the performance of A-LSTM when the GloVe embedding dimensions are 50, 100, 200, and 300. It can be seen from the figure that the overall performance of A-LSTM gradually increases with the increase of the GloVe embedding dimension. This shows that A-LSTM performs better when using words with richer meanings. Fig. 8 shows the performance of A-LSTM when the review representation dimension is set to 64, 128, 256, 512, and 768. We can see that the review representation dimension has a small impact on the performance of A-LSTM. When the dimension is 512, A-LSTM has the highest accuracy and $F1$, which are 90.9% and 88.3%, respectively.

D. Ablation Study

To confirm the effectiveness of the attention gate and attention layer designed in A-LSTM, we designed the following ablation experiments. We remove the attention gate, remove the attention layer, and remove both the attention gate

TABLE III
PERFORMANCE COMPARISON OF REMOVING A
CERTAIN A-LSTM STRUCTURE

	ACC	P	R	F1
LSTM	78.6	75.7	79.4	77.5
LSTM + attention gate	86.4	87.3	84.8	86.0
LSTM + attention layer	80.5	78.4	83.6	80.9
A-LSTM	90.9	88.7	87.9	88.3

Note: Bold entries indicate the best value.

TABLE IV
A-LSTM PERFORMANCE USING DIFFERENT
WORD REPRESENTATION

Word Representation	ACC	P	R	F1
GloVe embedding	88.2	84.6	84.5	84.5
GloVe + NER embedding	88.7	87.3	86.5	86.9
GloVe + TF-IDF embedding	89.1	86.5	87.2	86.8
GloVe + POS embedding	88.5	85.9	86.8	86.3
Mixed embedding	90.9	88.7	87.9	88.3

Note: Bold entries indicate the best value.

and attention layer to detect fake reviews in A-LSTM, respectively, and the experimental results are shown in the following Table III. Removing the attention gate and attention layer, the A-LSTM degenerates into a plain LSTM. As we can see in Table III, when the attention gate and attention layer are removed, the detection effectiveness of plain LSTM decreases dramatically compared with A-LSTM. Accuracy decreases by 12.3%, precision by 13%, recall by 8.5%, and *F1* by 10.8%. Introducing the attention gate to the plain LSTM, the detection effect is greatly improved. The introduction of attention gates to the plain LSTM resulted in a tremendous improvement in detection. Accuracy rose by 7.8%, precision by 11.6%, recall by 5.4%, and *F1* by 8.5%. It can be seen that the introduction of attention gates helps the model to fully learn the importance of individual words under the global semantics and fully encode the semantic information of the words. When the attention layer is introduced into the plain LSTM, the detection effect is also improved to a certain extent, but the improvement is not as large as the introduction of the attention gate. Accuracy rose by 1.9%, precision by 2.7%, recall by 4.2%, and *F1* by 3.4%. This shows that the attention layer is useful for encoding the semantics of fake reviews, which learns distant dependencies between words and further encodes the semantics of the reviews.

To assess the benefits of introducing NER embedding, TF-IDF embedding, and POS embedding in capturing word information, we compared the performance of A-LSTM in detecting fake reviews using different embedding combinations. As shown in Table IV, GloVe embedding performs comparably in accuracy to other embeddings but slightly lower in precision, recall, and *F1* value. This indicates that GloVe embedding provides a solid foundation for performance but can be further improved by adding other features. The inclusion of NER embedding leads to improvements in accuracy, precision, recall, and *F1* value, especially in precision and recall, demonstrating that NER embedding helps capture named entity information in

TABLE V
IMPACT OF LSTM LAYER NUMBERS ON
A-LSTM PERFORMANCE

	F1	64	128	512	256	768
1	76.3	77.5	79.2	81.5	82.3	
2	80.5	82.0	83.6	85.3	86.0	
3	82.9	85.2	86.5	87.7	87.8	
4	84.7	86.3	87.8	88.3	88.1	
5	85.3	86.0	84.1	83.6	83.0	
6	85.0	84.5	82.9	80.9	80.5	

Note: Bold entries indicate the best value.

reviews and enhances model performance. Introducing TF-IDF embedding also enhances overall performance, particularly in accuracy and recall, indicating that TF-IDF embedding captures keyword frequency information in reviews and improves model recall. Although POS embedding performs slightly less than NER and TF-IDF, it still brings some improvement, especially in precision. The mixed embedding, combining GloVe embedding with NER, TF-IDF, and POS features, achieves the best performance across all evaluation metrics, particularly with the highest *F1* value, indicating that the introduction of mixed features comprehensively captures semantic and structural information in reviews, thereby enhancing model performance. By analyzing this data, we can conclude that NER, TF-IDF, and POS embeddings all effectively enhance the detection of fake reviews, and combining these features with GloVe embedding further improves model performance. Therefore, considering a combination of multiple features is an effective strategy in fake review detection tasks, improving the model's understanding and analysis of review content, and ultimately enhancing detection accuracy and efficiency.

From the Table V, we can observe the performance of A-LSTM in detecting fake reviews across different numbers of A-LSTM layers and various hidden state dimensions. Here's an analysis of how different hidden state dimensions and A-LSTM layer numbers affect the performance of A-LSTM:

- 1) Initially, as the number of A-LSTM layers increases, the performance of A-LSTM generally improves. This is evident in the increasing trend of *F1* scores from 1 to 4 layers across all hidden state dimensions. Beyond a certain point (in this case, for 5 and 6 layers), the performance starts to decline. This decrease in performance is likely due to issues such as gradient vanishing and overfitting, which become more pronounced as the model becomes deeper. The peak performance is achieved at 4 A-LSTM layers for all hidden state dimensions, indicating that this configuration strikes the optimal balance between feature learning and generalization.
- 2) The table suggests that as the hidden state dimension increases, the model's capacity to learn complex patterns improves. This is reflected in the generally higher *F1* scores for higher dimensional hidden states across different numbers of A-LSTM layers. However, higher dimensional hidden states also increase the risk of overfitting, as evidenced by the decline in performance for certain configurations with higher dimensions (e.g., 768).

- 3) For a hidden state dimension of 64, the A-LSTM with 5 layers exhibits the strongest feature learning ability, suggesting that in this scenario, increasing the number of layers enhances the model's capacity to capture intricate patterns. On the other hand, for other hidden state dimensions, the A-LSTM with 4 layers demonstrates the strongest learning ability, indicating that a balance between depth and complexity is crucial for optimal performance. The observation that higher hidden state dimensions correspond to stronger learning capabilities aligns with the understanding that larger dimensions allow the model to represent more intricate relationships within the data. However, this comes with a tradeoff, as higher dimensions also increase the risk of overfitting.

In conclusion, regardless of the hidden state dimension, increasing the number of A-LSTM layers initially enhances the performance of A-LSTM by improving feature learning and pattern capturing abilities. However, this improvement is followed by a decline in performance due to issues such as gradient vanishing and overfitting. The optimal configuration for A-LSTM, in terms of balancing depth and complexity, is observed at four LSTM layers, showcasing the critical interplay between model depth, hidden state dimension, and performance.

To demonstrate the effectiveness of our proposed attention loss function, we compare it with the fixed weight cross-entropy loss, focal loss, and poly loss. In the attention loss function, we consider giving appropriate attention to nonoutlier hard-to-classify samples to solve the sample imbalance problem. When λ is 0.05, the maximum attention multiplier is 10, which approximates the ratio of the number of complete samples to the number of nonoutlier hard-to-classify samples. We set μ to 0.9. A sample is considered an outlier when its hard-to-classify degree exceeds 0.9. The fixed weight cross-entropy loss function addresses class imbalance by assigning different weights to different classes. Typically, since the proportion of fake reviews is lower, higher weights are assigned to the fake review class to increase the model's focus on detecting fake reviews. In experiments, the ratio of weights between normal reviews and fake reviews is determined by the ratio of fake reviews to normal reviews in the test set. Focal loss was proposed by Lin et al. [61] to improve the overall performance of the model by weighting the loss function according to the degree of difficulty in classifying the samples. The focal loss gives more weight to samples that are difficult to classify and less weight to samples that are easy to classify. PolyLoss, proposed by Leng et al. [62], is used to address class imbalance issues in tasks such as image segmentation. By employing a polynomial-based weight adjustment mechanism, it places greater emphasis on classes with lower frequencies, effectively enhancing the model's ability to learn from minority classes. Our attention loss function defines a threshold, and when the difficulty of classification exceeds the threshold, its weight is reduced to avoid the model being misled by outliers. As can be seen from Table VI, the use of the attention loss function enables better training of the model than the other loss function. The fixed weight cross-entropy loss function assigns fixed weights, which cannot adaptively fit the importance of each class, resulting

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT
LOSS FUNCTIONS

	ACC	P	R	F1
Fixed weight CE	85.2	82.3	84.7	83.5
Focal loss	86.7	83.4	87.8	85.5
Poly loss	86.8	83.8	87.5	85.6
Attention loss	90.9	88.7	87.9	88.3

Note: Bold entries indicate the best value.

TABLE VII
PERFORMANCE COMPARISON BETWEEN LINGUISTIC FEATURES,
BEHAVIORAL FEATURES, AND THEIR COMBINATION

	ACC	P	R	F1
Review representation	82.4	82.8	83.1	82.9
Behavioral features	71.3	69.8	72.0	70.9
Review representation + behavioral feature	90.9	88.7	87.9	88.3

Note: Bold entries indicate the best value.

in the poorest model performance. Both focal loss and poly loss employ mechanisms to dynamically adjust weights, enabling the model to better learn features of minority classes. However, Focal Loss and poly loss tend to assign significant weights to outlier data in the dataset, leading to overfitting on these outliers. In contrast, the attention loss function can appropriately reduce the weights of outlier data, resulting in the best performance.

To analyze the respective roles of linguistic and behavioral features, we compared the performance of detecting fake reviews using linguistic features and behavioral features separately. From Table VII, it can be observed that linguistic features exhibit relatively good performance, with accuracy, precision, recall, and *F1* value of 82.4%, 82.8%, 83.1%, and 82.9%, respectively. This indicates that linguistic features effectively capture the semantic information of reviews. Similarly, when using behavioral features alone for detecting fake reviews, decent performance is achieved, with accuracy, precision, recall, and *F1* value of 71.3%, 69.8%, 72.0%, and 70.9%, respectively. Despite covering only the behavioral aspects of reviews, behavioral features still provide clues to assist in fake review detection, demonstrating some detection capability. When linguistic features are combined with behavioral features, there is a significant improvement in the detection performance of fake reviews. This suggests that combining linguistic and behavioral features can leverage the complementary nature of the two, thereby enhancing the accuracy and efficiency of fake review detection. In summary, while linguistic features effectively capture a lot of semantic information from review text, their information is limited. Behavioral features alone also demonstrate some effectiveness in detecting fake reviews. However, combining linguistic and behavioral features can achieve better performance in fake review detection. This underscores the importance of utilizing combined features to fully leverage the complementary nature of textual and behavioral information, thereby enhancing the detection capability of the model.

E. Comparison

In order to demonstrate the performance of our model, we compare our method with other baseline models and transformer-based deceptive review detection models. The compared models are as follows:

- 1) GAS [19] is a fraud detection model based on GCN, which combines homogeneous and heterogeneous graphs to mine semantic information. In the experiments, DeBERTa is used to embed review texts as initial features of nodes. The method trains 8 epochs with a maximum number of samples of 16 for the complete graph and 8 for the comment graph.
- 2) MGSD [20] is an improved multiiterative graph-based model that captures the relationships between neighbors during the iteration process. The features used for initial spamicity calculation in the experiments include rating deviation, bursty review behavior, bursty time window, etc. The sample size is set to 100 and the number of iterations is set to 10. The algorithm will stop early when the difference between the current and previous spamicity scores of all samples is less than 0.05.
- 3) BiLSTMWF [10] introduces part-of-speech features and first-person pronoun features into word representations, uses BiLSTM to encode document-level representations of reviews, and finally combines behavioral features to detect fake reviews. In the experiments, the dimension of Glove embedding is 300, the dimension of part-of-speech features is 32, and the dimension of first-person pronoun features is 20. Bi-LSTM with two-layer, 512-D hidden layer states is used. The batch size is set to 32 and the learning rate is set to 0.0001.
- 4) DFFNN [15] uses a deep feed-forward network and combines bag-of-words features, word embeddings, and sentiment features to learn review representations. The hidden layer dimension is set to 100, the input layer dropout rate is 0.2, the hidden layer dropout rate is 0.5, b is 100, η is 0.1, and T is 1000. The batch size is set to 16 and the learning rate is set to 0.00003.
- 5) ScoreGAN [22] is an improved GAN, where LSTM is the generator that produces hidden states of reviews based on scores, and CNN is the discriminator that generates reviews and detects their deception. The word embedding dimension is 200, the mapped feature dimension is 200, the CNN sliding window size is 3, the batch size is set to 64, and the learning rate is 0.0001.
- 6) ELECTRA [57] is an encoder with a pretraining task based on replaced token detection. In the experiments, [cls] token embeddings are used as review text representations and combined with behavioral features to detect fake reviews. The hidden size is 256, the batch size is 16, and the learning rate is 0.00005.
- 7) DeBERTa [58] is a novel BERT improvement based on a disentangled attention mechanism. In the experiments, [cls] token embeddings are used as review text representations and combined with behavioral features to detect fake reviews. The hidden size is 768, the batch size is 8, and the learning rate is 0.00005.
- 8) A-LSTM is an attention-based LSTM that we propose. The dimension of Glove embedding is 300, the dimension of POS embedding is 32, the dimension of TF-IDF embedding is 1, and the dimension of NER embedding is 10. The hidden layer dimension and review representation dimension are 512. λ is 0.05 and μ is 0.9. The batch size is 64 and the learning rate is 0.0001.

TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT METHODS

	ACC	P	R	F1
GAS	82.4	82.8	83.1	82.9
MGSD	83.6	83.6	84.3	83.9
BiLSTMWF	87.5	84.9	85.4	85.1
DFFNN	87.1	82.4	83.3	82.8
ScoreGAN	86.7	85.6	84.9	85.2
ELECTRA	89.4	88.2	87.6	87.9
DeBERTa	89.8	88.4	87.8	88.1
A-LSTM	90.9	88.7	87.9	88.3

Note: Bold entries indicate the best value.

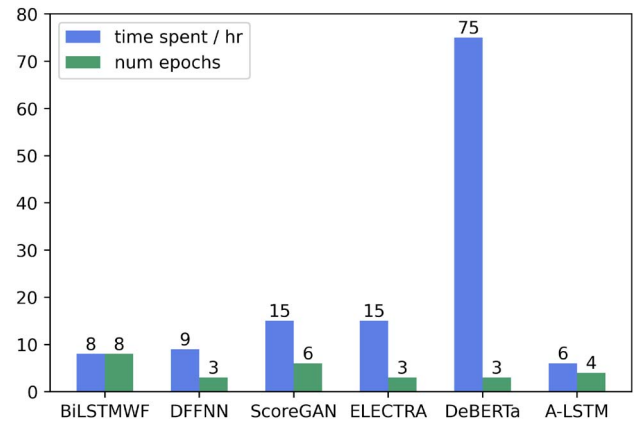


Fig. 9. Training time and number of epochs for different methods.

The result comparison is shown in Table VIII. Compared with other methods, A-LSTM has the best performance. Its accuracy reaches 90.9%, precision reaches 88.7%, recall reaches 87.9%, and $F1$ reaches 88.3%. GAS and MGSD are transductive graph-based methods that are good at mining the relationships between nodes, but have shortcomings in semantic mining of review content and use of behavioral features, resulting in limited detection performance. BiLSTMWF has improved in mining the semantics of comment content and integrating behavioral features. Compared with A-LSTM, BiLSTMWF does not mine the importance of words, so it performs worse than A-LSTM. Although DFFNN does an effective fusion of features, it is obviously less effective than A-LSTM in discovering the global semantics. ScoreGAN employs LSTM to encode the review text and CNN to detect the fraudulence of the review representation. However, LSTM is inferior to A-LSTM in capturing the semantic features of the text, resulting in a lower performance of ScoreGAN compared to A-LSTM detection. The results show that the transformer-based methods ELECTRA and DeBERTa perform closely behind A-LSTM, with an accuracy of about 1% lower than A-LSTM,

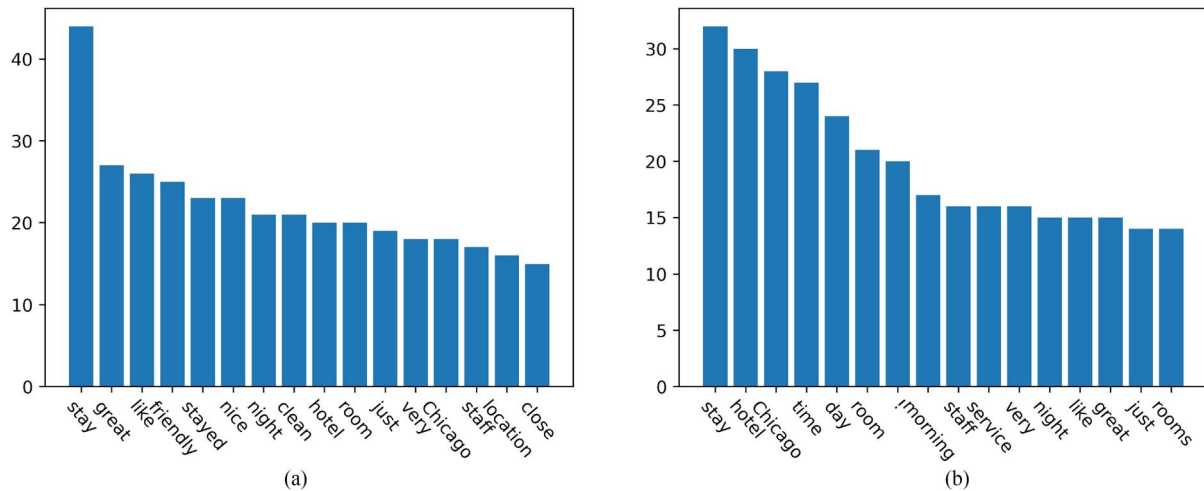


Fig. 10. Comparison of word frequency of high-weight words of real reviews and fake reviews in YelpChi hotel dataset. These words represent the most commonly occurring terms in both real and fake reviews from the YelpChi hotel dataset, offering valuable insights into customer sentiments and experiences. (a) 16 high-weight words with the highest frequency in real reviews of the YelpChi hotel dataset. (b) 16 high-weight words with the highest frequency in fake reviews of the YelpChi hotel dataset.

and precision, recall, and *F1* almost catching up with A-LSTM. The reason for this is that the pretrained ELECTRA and DeBERTa possess strong semantic modeling abilities, which enable them to capture the semantic features of the review text effectively. However, it should be noted that the transformer-based models fine-tuned on the fake review detection task suffer from issues such as large parameter size, challenging training, and low interpretability. The proposed A-LSTM can outperform them with fewer parameters, which fully validates the effectiveness of the proposed method.

Fig. 9 shows the training time and training epochs for A-LSTM and comparison methods. They are the hours and epochs required for the method to perform at its best, respectively. As GAS and MGSD are transductive graph-based methods, they cannot apply the existing training process to unknown data and need to relearn whenever the data changes and they do not participate in the comparison. As can be seen from Fig. 9, ELECTRA and DeBERTa are the most time-consuming models to fine-tune per epoch on average, taking 5 hours and 15 hours, respectively. The two models have about 100 million and 500 million parameters, respectively. However, due to their extensive pretraining on large-scale datasets, they can fully encode the semantic information of review text with only 3 epochs. BiLSTMWF has about 3 million parameters and a fast average training speed per epoch, but it requires more epochs to fit the data. A-LSTM enhances LSTM by introducing attention gates and attention pooling layers, which improve its ability to mine word information and enable faster data fitting. DFFNN has a slow training speed due to its tedious word preprocessing features. ScoreGAN encodes text representation with LSTM and classifies with CNN, which is more complex and time-consuming than A-LSTM. In summary, A-LSTM outperforms other methods in terms of efficiency.

All in all, A-LSTM is an attention LSTM model specifically designed to address the characteristics of fake reviews containing a large amount of similar content and a small number

of words embodying more deception, capturing information about the importance of each word to the overall semantic representation. Existing methods of detecting fake reviews have achieved good results on their targeted research problems, but they have shortcomings in embedding review texts and discovering fraudulent content in reviews. Although transformer-based methods can achieve good results, they face problems of large parameter size and difficulty in training. A-LSTM can achieve better detection performance with fewer resources. Therefore, the review embedding method proposed in this article surpasses other state-of-the-art methods.

F. High-Weighted Words Comparison Analysis

The attention gate of A-LSTM calculates the weight of each word when encoding global semantics. The higher the weight of a word, the stronger its importance. In order to observe the difference between high-weight words of real reviews and fake reviews, we take the top eight words (Top8) by weight for each review and then perform word frequency statistics on Top8 of real reviews and fake reviews. Finally, we compare and analyze 16 words with the highest frequency in real reviews and fake reviews. Considering that function words such as “the,” “a,” “to,” and “I” do not have complete lexical meaning and punctuation marks such as “,” and “.” do not have actual meaning, this experiment filters out function words and punctuation marks without actual meaning, retaining meaningful words and common ones used to express emotions “!” and “?”. Figs. 10 and 11 show the word frequency comparison of YelpChi hotel data and YelpChi restaurant data, respectively. Since the number of real reviews is much larger than that of fake reviews, we performed random naive downsampling to make the number of real reviews consistent with that of fake reviews.

Fig. 10 shows the word frequency comparison of high-weight words in real and fake reviews of YelpChi hotel data. As can be seen from the figure, among the top 16 words in terms of

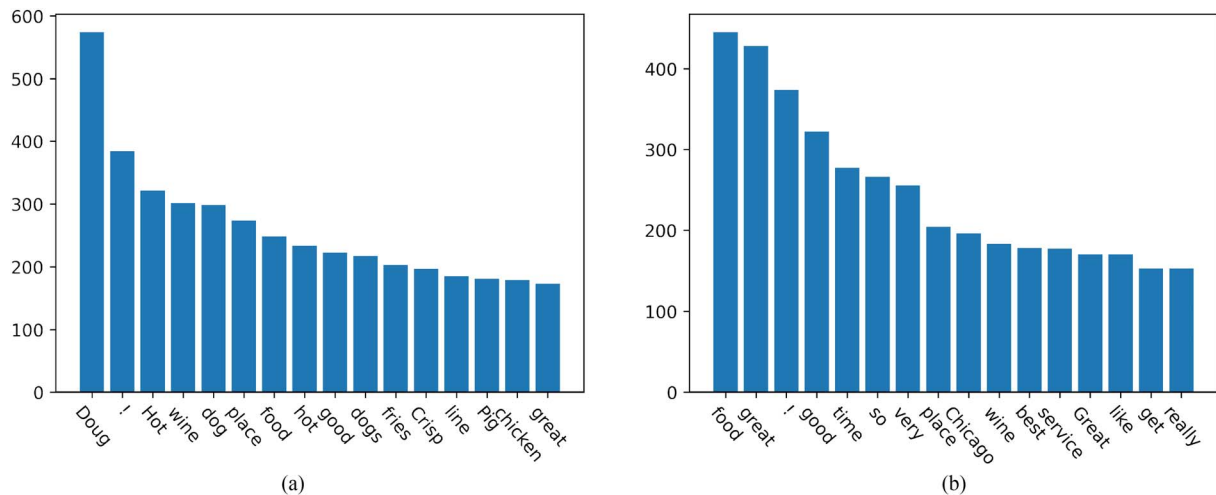


Fig. 11. Comparison of word frequency of high-weight words of real reviews and fake reviews in YelpChi restaurant dataset. These words represent the most commonly occurring terms in both real and fake reviews from the YelpChi restaurant dataset, offering valuable insights into customer sentiments and experiences. (a) 16 high-weight words with the highest frequency in real reviews of the YelpChi restaurant dataset. (b) 16 high-weight words with the highest frequency in fake reviews of the YelpChi restaurant dataset.

frequency, “stay,” “great,” “like,” “night,” “hotel,” “room,” “just,” “very,” “Chicago,” and “staff” appear in both real and fake review data, indicating that these words are basic vocabulary for reviews in the Chicago hotel domain; “friendly,” “nice,” “clean,” “location,” and “close” appear in the real review dataset, but not in the fake review dataset. These words can describe the hotel staff service, hotel hygiene condition, and hotel location. This shows that real reviews are very comprehensive in evaluating hotels, covering various dimensions. These words help to find real reviews in the hotel dataset. “Time,” “day,” “!,” “morning,” and “service” appear in the fake review dataset, but not in the real review dataset. Among these words, “!” is used to express a surprised tone, while other words are used to describe time and service. It can be inferred that fake reviews have a stronger tone, and because fake reviewers do not have a real consumption experience, their evaluation angle is more one-sided.

Fig. 11 shows the word frequency comparison of high-weight words in real and fake reviews of YelpChi restaurant data. As can be seen from the figure, the difference in frequency of “!” in the Chicago restaurant dataset is not much, which indicates that there is not much difference in tone between real and fake reviews in restaurant data. “Wine,” “food,” “good,” and “great” appear in both real and fake reviews, as basic vocabulary for reviews in the restaurant domain. The real review dataset contains “Doug,” “Hot,” “dog,” “dogs,” “Crisp,” and “Pig.” These restaurant names, but not in fake reviews, which can be inferred that real reviewers tend to mention restaurant names when evaluating. The real review dataset contains words that do not appear in fake reviews such as “fries,” “line,” and “chicken,” which can express details such as dishes consumed and queuing experience. Only words that appear in the fake review dataset are “so,” “very,” “best,” “like,” and “really” for expressing emotions, while service and time involve service and time, which can be inferred that fake reviews tend to express strong positive emotions on service and time aspects.

V. CONCLUSION

In the deceptive review detection task, the representation of the review text is crucial to the detection performance. The embedding work of review text serves as the basis for some fake review detection methods, which directly determines the performance of the method in identifying fake reviews. However, the existing research has many deficiencies in the textual representation of deceptive reviews. For example, feature engineering is cumbersome and methods are too heavy. In this article, we propose an efficient and lightweight A-LSTM method to learn the representation of review text for the problem of deceptive review detection. A-LSTM introduces an attention gate and attention layer and discovers word importance information, making review representation more effective in fake review detection tasks. Moreover, because deceptive review detection is a binary classification task with unbalanced sample categories, model training will be biased. We propose attention loss to solve this problem, it pays more attention to those difficult-to-classify samples to balance the bias caused by the imbalance of sample categories. Through comprehensive experiments on the Yelp review dataset from the real world, compared with other models, our model achieves state-of-the-art performance on deceptive review detection tasks. By counting the frequency of the most important words in the data set, we analyzed the characteristics of fake and real reviews of Yelp Chicago restaurants and hotels, as well as the writing techniques of fake reviewers.

The proposed lightweight review text embedding method, A-LSTM, outperforms other state-of-the-art methods in the detection of fake reviews and outperforms transformer-based methods with a huge number of parameters, which fully demonstrates the effectiveness of the proposed method. The calculation of the importance of words by A-LSTM is very meaningful when used to analyze the characteristics of fake reviews. However, the method still has some shortcomings. This

article combines behavioral features to improve the detection performance when detecting fake reviews. However, new users do not have enough data to construct behavioral features. Therefore, the reviews of new users will hinder the further improvement of the detection effect. Future research can consider the lack of features for new users and conduct research to address the lack of behavioral features to further improve detection.

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